Sri Chaitanya IIT Academy., India.

AP, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office, Madhapur – Hyderabad

Sec: Sr-ICON-B-1	**JEE-ADVANCE-2018-P2**	**Date:** 09-05-21
Time: 02:30 PM to 05:30 PM	**GTA-03**	**Max.Marks:**180

SYLLABUS:

Physics : TOTAL SYLLABUS

Chemistry : TOTAL SYLLABUS

Mathematics : TOTAL SYLLABUS

Name of the Student: ____________________ **H.T. NO:**

JEE-ADVANCE-2018-P2-Model

Time: 3h **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 7 – 14)	Questions with NV Integer Answer Type	+3	0	8	24
Sec – II(Q.N : 15-18)	Matrix Matching Type	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 25 –32)	Questions with NV Integer Answer Type	+3	0	8	24
Sec – II(Q.N : 33-36)	Matrix Matching Type	+3	-1	4	12
Total				**18**	**60**

MATHEMATICS:

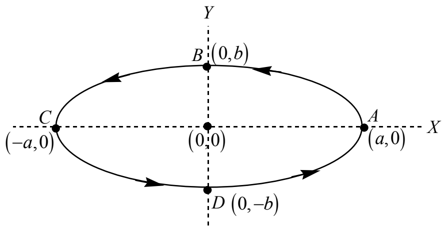
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N:37 – 42)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N :43–50)	Questions with NV Integer Answer Type	+3	0	8	24
Sec – II(Q.N : 51-54)	Matrix Matching Type	+3	-1	4	12
Total				**18**	**60**

PHYSICS

SECTION 1

This section contains **SIX (06)** questions. Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question. Answer to each question will be evaluated according to the following marking scheme: *Full Marks*: +4 If only (all) the correct option(s) is (are) chosen. *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen. *Partial Marks*: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options. *Partial Marks* +1 If two or more options are correct but ONLY one option is chosen and it is acorrect option. *Zero Marks*: **0** If none of the options is chosen (i.e. the question is unanswered). *Negative Marks*: -2 In all other cases. **For Example:** If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options),without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option) ,without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A particle of mass 'm' is taken with a constant speed V_0 along the circumference of an elliptical orbit with a semi major axis of length 'a' and semi minor axis of length 'b' as shown in fig. then which of the following statements are correct :



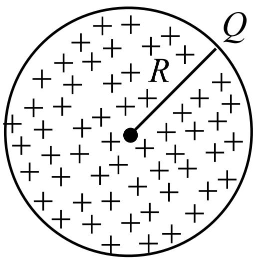
A) Radius of curvature of the elliptical orbit at (-a, 0) is $\frac{b^2}{a}$

B) Radius of curvature of the elliptical orbit at (0, -b) is $\sqrt{\frac{a^2+b^2}{2}}$

C) The magnitude of angular momentum of the particle when it crosses the (0, b) about the origin is mV_0b

D) The magnitude of acceleration of the particle when it crosses point (0, -b) is $\frac{V_0^2 b}{a^2}$

2. A Non-conducting solid sphere of radius 'R' and charge '+Q' is distributed uniformly throughout the volume, as shown in fig. (Relative permittivity inside and out side the Non-conducting sphere is taken as one) then which of the following statements are correct.



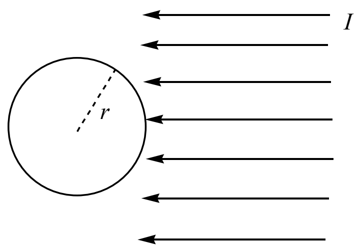
A) The electro-static potential energy in the form of the electric field with in the Non-conducting solid sphere is $\frac{Q^2}{40\pi\varepsilon_0 R}$

B) The electro –static potential energy in the form of the electric field out side the non conducting solid sphere (from the surface to infinity) is $\frac{Q^2}{8\pi\varepsilon_0 R}$

C) The self energy of the non-conducing charged solid sphere (neglect gravitational self energy) $\frac{3Q^2}{20\pi\varepsilon_0 R}$

D) Electro static potential of the non-conducting sphere at it's centre (infinity is taken as zero reference for the electric potential) is $\frac{3Q}{8\pi\varepsilon_0 R}$

3. A uniform solid sphere of radius 'r' is kept in the path of a parallel beam of light large aperture. If the parallel beam intensity 'I' as shown in fig. : (velocity of light 'C') then which of the following statements are correct?



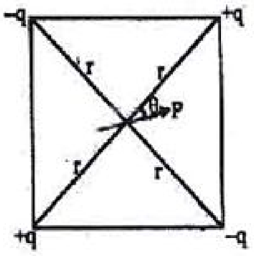
A) If the solid sphere material is perfectly absorbing then the net force exerted by the light beam on that sphere is $\frac{I\pi r^2}{C}$

B) If the solid sphere material is perfectly reflecting surface then the net force exerted by the beam on the solid sphere is $\frac{2I\pi r^2}{C}$

C) If the absorption coefficient, reflection coefficient of the solid sphere material is 0.3 and 0.7 respectively then the net force exerted by the beam on that solid sphere is $\frac{I\pi r^2}{C}$

D) If the absorption coefficient, reflection coefficient of the solid sphere material is 0.5 and 0.5 respectively then the net force exerted by the light beam on that solid sphere is $\frac{1.5I\pi r^2}{C}$

4. Given square frame of diagonal length 2r made of insulating wires. There is a short ideal electric dipole, having dipole moment P, fixed in the plane of the figure lying at the centre of the square, making an angle θ as shown in figure. Four identical particles having charges of magnitude q each and alternatively positive and negative are placed at the four corners of the square. Select the correct alternative(s). (neglect gravitational effects) $\left(k=\frac{1}{4\pi\varepsilon_0}\right)$ (Centre of dipole coincidence with the intersecting point of diagonals)



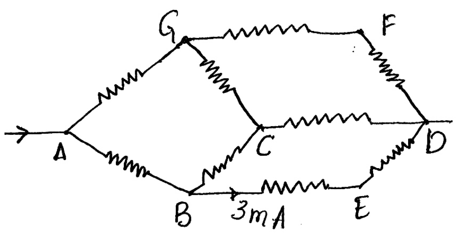
A) Resultant electrostatic force on the system of four charges due to dipole is $\frac{6kPq}{r^3}$

B) Resultant electrostatic force on the system of four charges due to dipole is $\frac{6kPq}{r^3}\cos\theta$

C) Net torque on the system of four charges about the centre of the square due to dipole is zero

D) Net torque on the system of four charges about the centre of the square due to dipole is $\frac{3kPq}{r^2}$

5. Consider nine identical resistors arranged as shown in figure. In this arrangement, electric current enters at node A and leaves from node D. Let $V_{AD}=5V$ and $I_{BE}=3mA$. Therefore



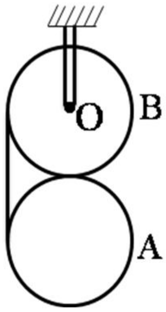
A) $I_{AB} = 5mA$

B) $I_{AB} = 10mA$

C) Effective resistance between A and D is 500Ω

D) Effective resistance between A and D is $\frac{500}{3}\Omega$

6. Two identical cylinders A and B are wrapped by a common ideal string. The cylinder B can rotate without friction about a horizontal axis passing through 'O'. Both the cylinders are confined to move in the vertical plane. Initially, the distance between centers of mass of both the cylinders is just greater than 2r ;where r is radius of each cylinder .If the system is released form rest, then (assume no slip anywhere)



A) Angular acceleration of A is $\frac{2g}{5r}$

B) Angular acceleration of B is $\frac{3g}{5r}$

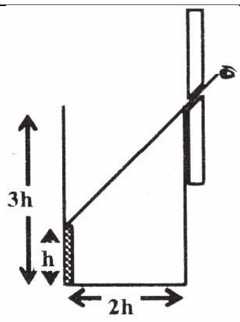
C) Acceleration of centre of mass of A is $\frac{4g}{5}$

D) Distance covered by centre of mass of A at time 't' after the release is $\frac{2}{5}gt^2$

SECTION 2

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded -off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, - 127.30) using the mouse and the on- screen virtual numeric keypad in the place designated to enter the answer. Answer to each question will be evaluated according to the following marking scheme: *Full Marks*: +3 If ONLY the correct numerical value is entered as answer. *Zero Marks* : 0 In all other cases.

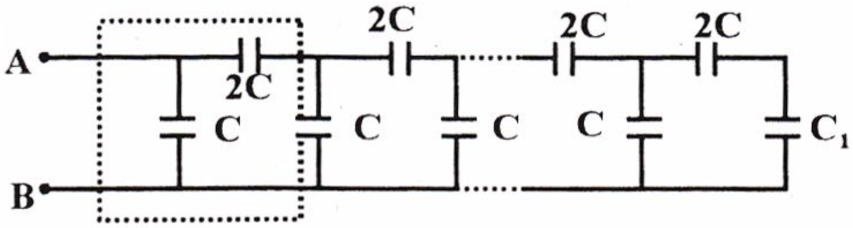
7. An observer can see through a pin-hole the top end of a thin rod of height h, placed as shown in the figure. The beaker height is 3h and its radius h. When the beaker is filled with a liquid up to a height 2h, he can see the lower end of the rod. Then the refractive index of the liquid is



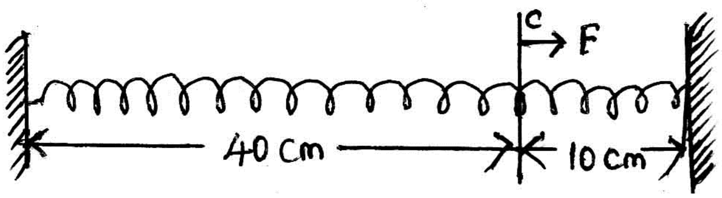
8. If a block is pulled up an inclined plane with a force of 10 N along the line of greatest slope, the block sides up with a constant velocity. If the block is pulled down the inclined plane with a force of 2N along the line of greatest slope, the block slides down with a constant speed. Find the force of kinetic friction between the block and the plane in 'newton'.

9. A car blowing a horn of frequency 350 Hz is moving normally towards a wall with a speed of 5 m/s. velocity of sound in air is 350 m/s . If the beat frequency heard by a person standing between the car and the wall is (2n) Hz, find the value of n.

10. The figure shows a network of capacitance consisting of several repetitive units shown by doted square and a capacitor of capacitance C_1 connected at end as shown in the figure. The value of C_1 such that the equivalent capacitance between A and B is independent of the number of units is ___C

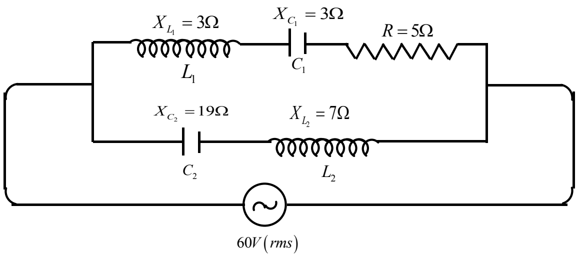


11. A light uniform spring of force constant 1N/cm is connected between two walls so that the spring is just taut. Point 'C' is located on the spring at distance 40 cm from the left end. Now a force of F=5N is exerted along the spring at point 'C' as shown. Find the displacement of point 'C' in 'mm' from initial state to final equilibrium state.



12. The peripheral velocity of a particle located on the equator of a planet is V_0 due to rotation of the planet about its own axis. The apparent weight of the particle is half its true weight. If escape velocity of a polar particle from this planet is kv_0, find the value of k.

13. In the given A.C circuit the r.m.s voltage of the source is 60V then the r.m.s current passing through the source is___A

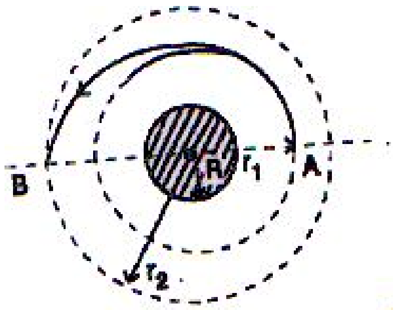


14. A calorimeter contains ice. $Q_1 = 500\,\text{Cal}$ of heat is required to heat it together with its contents from 270 K to 272 K and $Q_2 = 16{,}600$ Cal from 272 K to 274 K. If heat capacity of the calorimeter is ____ Cal/C^0. $L_{ice} = 80$ Cal/g. Specific heat of ice $= 0.5\ \text{Cal.g}^{-1}.\left(C^0\right)^{-1}$ and that of water $= 1\ \text{Cal.g}^{-1}\left(C^0\right)^{-1}$. System is at normal atmospheric pressure.

SECTION 3

This section contains **FOUR (04)** questions. Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**. **FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching. For each question, marks will be awarded according to the following marking scheme: *Full Marks*: +3 If ONLY the option corresponding to the correct matching is chosen. *Zero Marks*:0 If none of the options is chosen (i.e. the question is unanswered). *Negative Marks*: -1 In all other cases.

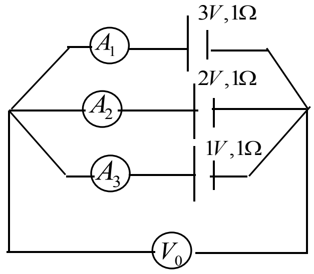
15. A space vehicle moving in a circular orbit of radius r_1 transfer to a larger circular orbit of radius r_2 by means of an elliptical path between A and B. The transfer is accomplished by a burst of speed ΔV_A at A and a second burst of speed ΔV_B at B. Then match the following : (R is the radius of the earth, g is the acceleration due to gravity on the surface of the earth, neglect the spin rotation of the earth)



Column I		Column II	
p)	Velocity at A in a circular orbit of radius r_1	1)	$R\sqrt{\frac{2g\,r_2}{r_1(r_1+r_2)}}$
q)	Velocity at A of elliptical orbit (perihelion (near point) of the transferred orbit)	2)	$R\sqrt{\frac{2g\,r_1}{r_2(r_1+r_2)}}$
r)	Velocity at B of elliptical orbit (aphelion (far point) of the transferred orbit)	3)	$R\sqrt{\frac{g}{r_2}}\left(1-\sqrt{\frac{2r_1}{r_1+r_2}}\right)$
s)	Velocity increment at B from elliptical path to circular orbit	4)	$R\sqrt{\frac{g}{r_1}}$

A) p-4, q-1, r-2, s-3 **B)** p-4, q-2, r-1, s-3
C) p-4, q-1, r-3, s-2 **D)** p-4, q-3, r-2, s-1

16. In the given circuit Ammeters and voltmeter are ideal and their numerical values. In S.I. system as shown in the column II



Column I		Column II	
p)	The reading of ideal Ammeter A_1 is	1)	1
q)	The reading of ideal Ammeter A_2 is	2)	0
r)	The reading of ideal Ammeter A_3 is	3)	2
s)	The reading of ideal Voltmeter V_0 is	4)	3

A) p-1, q-2, r-2, s-3 **B)** p-1, q-2, r-1, s-3
C) p-1, q-1, r-2, s-1 **D)** p-2, q-1, r-1, s-3

17. A neutron moving with kinetic energy K collides with a hydrogen atom at rest in ground state. Assume mass of neutron to be equal to that of hydrogen atom. Match the kinetic energy of neutron with the possible phenomena mentioned in column II. Energy of hydrogen in ground state is –13.6 ev

Column I (Kinetic energy of colliding neutron)		Column II	
p)	K = 10.2 eV	1)	Elastic collision may take place
q)	K = 13.6 ev	2)	Inelastic collision may take place
r)	K = 20.4 ev	3)	Perfectly inelastic collision may take place
s)	K = 22.0 ev	4)	Hydrogen atom may get ionised

A) p-1,q-1,r-12, s-13 **B)** p-1,q-4,r-3, s-2

C) p-1,q-1,r-13, s-12 **D)** p-1,q-4,r-12, s-2

18. In each situation of Column-I, and incident wavefront and its corresponding reflected or refracted wavefront is shown. In column-II the optical instrument used for reflection or refraction is given. Always take the optical instrument to the right of incident wavefront. The incident wavefront is moving towards right. Match each pair of incident and reflected/refracted wavefront in column-I with the correct optical instrument given in column-II

[Consider the shape of the refracted wave fronts in case of lens only between the second principal focus and lens, virtual wave fronts are also taken into consideration] same condition is applied in case of mirrors

Column I Incident Wavefront	Reflected/Refracted wavefront	Column II Optical instrument used	
p)		1)	
q)		2)	
r)		3)	
s)		4)	

A) p-2,4; q-1,3; r-1,4; s-2,3 **B)** p-1,3; q-2,4; r-2,3; s-1,4

C) p-2,3; q-1,3, r-2,4; s-1,4 **D)** p-2,4; q-1,3; r-2,3; s-1,4

CHEMISTRY

SECTION 1

This section contains **SIX (06)** questions. Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question. Answer to each question will be evaluated according to the following marking scheme: *Full Marks*: +4 If only (all) the correct option(s) is (are) chosen. *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen. *Partial Marks*: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options. *Partial Marks* +1 If two or more options are correct but ONLY one option is chosen and it is acorrect option. *Zero Marks*: **0** If none of the options is chosen (i.e. the question is unanswered). *Negative Marks*: -2 In all other cases. **For Example:** If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options),without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option) ,without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

19. A photon of wavelength is $4\times10^{-7}m$ strikes on a metal surface, the work function of the metal being $3.4\times10^{-19}J$. Select the correct statements.

A) The energy of photon is $4.97\times10^{-19}J$

B) The kinetic energy of the emission is $1.57\times10^{-19}J$

C) The kinetic energy of the emission is $0.98ev$

D) The velocity of photo electron is $5.87\times10^{5}ms^{-1}$

20. Which of the following can act as oxidizing as well as reducing agent?

A) NH_3 **B)** HNO_3 **C)** SO_2 **D)** HNO_2

21. Correct matching of minerals and their compositions

A) Siderite - Carbonate mineral - Iron metal

B) Cassiterite - Silicate mineral - Tin metal

C) Sphalerite - Sulphate mineral - Zinc metal

D) Magnetite - Oxide mineral - Iron metal

22. Which of the following statements are correct ?

A)Nitrous oxide is a neutral oxide.

B) Nitrogen trioxide is acidic

C)Dinitrogen pentoxide is a deliquescent solid.

D) Dinitrogen tetroxides is a mixed anhydride

23. Which of the following statements are true?

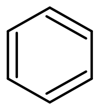
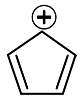
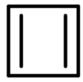
A) $CH_3-CH_2-CH_2-I$ will react more readily than $(CH_3)_2CHI$ for S_N2 reactions

B) $CH_3-CH_2-CH_2-Cl$ will react more readily than $CH_3-CH_2-CH_2-Br$ for S_N2 reactions

C) $CH_3-CH_2-CH_2-CH_2-OH$ will react more readily than $(CH_3)_3C-CH_2-OH$ for S_Ni reactions in the presence of pyridine

D) $CH_3-O-C_6H_5Br$ will react more readily than $NO_2-C_6H_5-CH_2Br$ for S_N2 reactions

24. Which of the following species is/are antiaromatic?

A)  B)  C)  D) CH_3^+

SECTION 2

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded -off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, - 127.30) using the mouse and the on- screen virtual numeric keypad in the place designated to enter the answer. Answer to each question will be evaluated according to the following marking scheme: *Full Marks*: +3 If ONLY the correct numerical value is entered as answer. *Zero Marks* : 0 In all other cases.

25. 1 mole of nitrogen is mixed with 3 moles of hydrogen in a $\sqrt{3}$ litre container where 66.67% of nitrogen is converted into ammonia by the following reation:

$N_2(g)+3H_2(g) \rightleftharpoons 2NH_3(g)$, then the value of K_C for the reaction

$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g)+\frac{3}{2}H_2$ will be..

26. An ideal gas expands against a constant external pressure of 2.0 atmosphere from 20 litre to 40 litre and absorbs 10 kJ of heat from surrounding. What is the change in internal energy of the system in joles? $\left(Given{:}1\ atm - litre = 101.3\ \ J\right)$

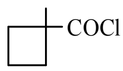
27. 'DO' value of water sample is $6\,ppm$. The weight of dissolved oxygen present in $100Kg$ of water sample in grams is:

28. When chloride salt is heated with acidified $K_2Cr_2O_7$ red vapours of X are liberated. Number of peroxy bonds in one molecule X are

29. One glucose molecule reacts with ' X ' molecules of phenyl hydrazine to form osazone. The value of 'X' is.

30. The sum of number of tetrahedral and trigonal planar units in borax is/are……

31. The number of compounds undergo H.V.Z reaction is

I. CH_3-COOH II. CH_3-CH(CH_3)-CHOOH III. Ph-CH_2-COOH IV. CH_3-COCl V. PhCOCl VI.  VII. CH_3-CH_2-COOH

32. The number of unshared pair of electrons in the electrophile involved in conversion of phenol to salicyladehyde using $CHCl_3$/KOH is

SECTION 3

This section contains **FOUR (04)** questions. Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**. **FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching. For each question, marks will be awarded according to the following marking scheme: *Full Marks*: +3 If ONLY the option corresponding to the correct matching is chosen. *Zero Marks*:0 If none of the options is chosen (i.e. the question is unanswered). *Negative Marks*: -1 In all other cases.

33. Match the following

Column I		Column II	
(a)	Linear shape	(p)	CS_2
(b)	Sp hybridization	(q)	N_3^-
(c)	sp^3d hybridization	(r)	I_3^-
(d)	CO_2 is isostructural to	(s)	NCO^-
		(t)	NO_2^+

A) a-pqrst; b-pqst; c-r; d-pqrst **B)** a-pqst; b-pqrt; c-r; d-qrst

C) a-qrst; b-pqs; c-s; d-pqrst **D)** a-pqrt; b-pqrst; c-q; d-pqrst

34. Match the following

Column – I	Column – II
A) CH_3^+	**p)** Electrically neutral
B) CH_3^-	**q)** having $6e^-$ in the outer shell
C) $CH_3^{\bullet}$	**r)** sp^2
D) $:CH_2$	**s)** sp^3

A) A-qr; B- s; C-pr; D-pqr **B)** A-qs; B- s; C-pq; D-pqr

C) A-q; B- qs; C-pr; D-qr **D)** A-r; B- pr; C-ps; D-qr

35. Match the following

Column – I		Column II	
A)	Cellulose	**p)**	Natural polymer
B)	Nylon-6, 6	**q)**	Synthetic polymer
C)	Protein	**r)**	Amide linkage
D)	Sucrose	**s)**	Glycoside linkage

A) A-qr; B- s; C-ps; D-pqr **B)** A-qs; B- s; C-pq; D-pqr
C) A-q; B- qs; C-pr; D-qr **D)** A-ps; B- qr; C-pr; D-s

36. Where $-t_{1/n}$ is the time during which concentration is reduced to 1/n times of the initial value

	Column I		Column II
A)	$t_{1/2}=\frac{0.693}{k}$	**P)**	Zero order
B)	$t_{1/2}=\frac{a}{2k}$	**Q)**	First order
C)	$\tau=\frac{1}{k}$	**R)**	Average life
D)	$t_{3/4}=2t_{1/2}$	**S)**	75% completion
		T)	second order

A) A-P; B- Q; C-SQ; D-R **B)** A-Q; B- P; C-R; D-SQ
C) A-R; B- S; C-Q; D-Q **D)** A-S; B- R; C-SQ; D-Q

MATHEMATICS

SECTION 1

This section contains **SIX (06)** questions. Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question. Answer to each question will be evaluated according to the following marking scheme: *Full Marks*: +4 If only (all) the correct option(s) is (are) chosen. *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen. *Partial Marks*: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options. *Partial Marks* +1 If two or more options are correct but ONLY one option is chosen and it is acorrect option. *Zero Marks*: **0** If none of the options is chosen (i.e. the question is unanswered). *Negative Marks*: -2 In all other cases. **For Example:** If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options),without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option) ,without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

37. A and B are events with non- zero probabilities. In the usual notation, it is found that $P(A|B) > P(B|A)$ *and* $P(AB) \neq 0$. Then which of the following can be true?

A) $P(A^C) < P(B^C)$

B) $P(A \cap B^C) < P(A^C \cap B)$

C) $P(A \cap B) < 2P(B)$

D)A and B are independent $\Rightarrow P(A \cap B) < P((A))^2$

38. For the function $y = f(x) = \dfrac{1}{x^2 + 4x + 5}$ which of the following statements) always hold good?

A) the curve y = $f(x)$ has no vertical asymptotes

B) If $\int 3f(x)dx = A\tan^{-1}(x+B) + C$ where A, B , C are real coefficients then A = B.

C) Area enclosed by y = 3 $f(x)$, the x-axis and the lines x = $\sqrt{3}-2$ and x = a where a is the x-coordinate of the turning point, is π

D) range of $f(x)$ is $(0, 1]$

39. The complex numbers satisfying the equation $(3z+1)(4z+1)(6z+1)(12z+1) = 2$ is /are

A) $\dfrac{\sqrt{33}-5}{24}$ **B)** $\dfrac{\sqrt{33}+5}{24}$ **C)** $\dfrac{-i\sqrt{23}-5}{24}$ **D)** $\dfrac{-i\sqrt{23}+5}{24}$

40. Let $f(x)=ax^2-b|x|$, where a and b are constants. Then at x = 0, $f(x)$ has

A) a maxima whenever a > 0, b < 0

B) a maxima whenever a > 0, b > 0

C) a minima whenever a > 0, b < 0

D) neither a maxima nor a minima whenever a > 0, b < 0

41. If $I_n=\int_0^{\frac{\pi}{2}}\cos^n x\cos nx\,dx,\ n\in N$, then

A) $I_9=2I_{10}$ **B)** $2I_{20}=I_{19}$ **C)** $I_{15}=\frac{1}{2^{15}}$ **D)** $I_{20}=\frac{1}{2^{19}}$

42. If $\theta\in R$, then the curves $\frac{x^2}{\sec^2\theta}+\frac{y^2}{\tan^2\theta}=1$ and $\frac{x^2}{\cos^2\theta}-\frac{y^2}{\sin^2\theta}=1$

A) are confocal

B) have same directrices

C) have same length of the latus rectum

D) intersect orthogonally

SECTION 2

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded -off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, - 127.30) using the mouse and the on- screen virtual numeric keypad in the place designated to enter the answer. Answer to each question will be evaluated according to the following marking scheme: *Full Marks*: +3 If ONLY the correct numerical value is entered as answer. *Zero Marks* : 0 In all other cases.

43. A bag contains 5 balls, in which everyone is either white or black. Three balls drawn and found that 2 are white and one is black. If another ball is drawn without replacing the above three, then the probability that it is white is $\frac{a}{b}$ where a and b are relatively prime integers, then $\frac{a+b}{b}$

44. If ω is a non–real cube root of unity and $f(x)=\sum_{k=0}^{8}A_kx^k$.

If $f(x)+f(\omega x)+f(\omega^2x) = n\left(A_0+A_nx^n+A_{2n}x^{2n}\right)$ then n must be equal to

45. Let $a_1=1, a_2=1+a_1, a_3=1+a_1a_2$, ..., $a_{n+1}=1+a_1a_2.....a_n$,

then $\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{a_3}+......upto\ \infty$ is equal to

46. If $\lim\limits_{x\to 0}\frac{\tan^{-1}x+ae^{2x}+b\sin x+c\ln(1-x)}{x^3(x^2+1)}$ is finite and a,b,c are real numbers. Then value of a + b is -2λ, find $|\lambda|$?

47. If S_n = 1.n+2.(n–1) + 3.(n–2) + ... + n.1 and $S_{25}=26\lambda$ *then* λ is

48. A curve $y=f(x)$ is passing through (2, 0) and slope of its tangent at any point (x,y) *is* $\frac{(x+1)^2+y-3}{x+1}$. Then the area bounded by the curve $y=f(x)$ and the x-axis is

49. The greatest value of $f(x)=\left(3-\sqrt{4-x^2}\right)^2+\left(1+\sqrt{4-x^2}\right)^3$ is A, then A/5 =

50. Let A = $\begin{bmatrix} l & m & n \\ p & q & r \\ 1 & 1 & 1 \end{bmatrix}$ and B = A^2.

If $(l-m)^2+(p-q)^2=9$, $(m-n)^2+(q-r)^2=16$, $(n-l)^2+(r-p)^2=25$, then the value of $\frac{|B|}{5}$ is

SECTION 3

This section contains **FOUR (04)** questions. Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**. **FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching. For each question, marks will be awarded according to the following marking scheme: *Full Marks*: +3 If ONLY the option corresponding to the correct matching is chosen. *Zero Marks*:0 If none of the options is chosen (i.e. the question is unanswered). *Negative Marks*: -1 In all other cases.

51. Match the following

	Column I		Column II
p)	The remainder when 23^{23} is divided by 53 is	**1)**	55
q)	The number of integral solutions of the equation $x+y+z+\mu=3,\ x\ge -2,\ y\ge,-1,\ z\ge,0,\mu\ge 1$ is	**2)**	30
r)	The coefficient of x^2y in the expansion of $(1+x+2y)^5$ is	**3)**	56
s)	If $C_r=\binom{10}{r}$ then $\sum_{r=1}^{10}\frac{r.C_r}{C_{r-1}}$ is	**4)**	60

A) p-1, q-2, r-3, s-4
B) p-2, q-3, r-4, s-1
C) p-3, q-4, r-1, s-2
D) p-4, q-1, r-2, s-3

52. Match the following

Column I		Column II	
p)	If W is an imaginary cube root of unity and $x_1, x_2, x_3 \in R$, $\sum_{r=1}^{3} \frac{1}{x_r + W} = 2W^2$, $\sum_{r=1}^{3} \frac{1}{x_r + W^2} = 2W$ then $\sum_{r=1}^{3} \frac{1}{x_r + 1}$ is	**1)**	$6\sqrt{13}$
q)	If z_1, z_2, z_3 be the vertices A, B, C respectively of ΔABC such that $\lvert z_1\rvert = \lvert z_2\rvert = \lvert z_3\rvert$ and $\lvert z_1 + z_2\rvert = \lvert z_1 - z_2\rvert$ then tan C is	**2)**	2
r)	Number of solutions of the equation $e^{2x} = x^2$	**3)**	1
s)	If $\bar{a}, \bar{b}$ be any two vectors of magnitudes 2, 3 respectively such that $\left\lvert 2\left(\bar{a} \times \bar{b}\right)\right\rvert + \left\lvert 3\left(\bar{a}.\bar{b}\right)\right\rvert = K$ then maximum value of K is	**4)**	3

A) p-2, q-3, r-4, s-1 **B)** p-2, q-4, r-3, s-1

C) p-2, q-3, r-1, s-4 **D)** p-3, q-2, r-4, s-1

53. Match the following

Column I		Column II	
p)	If $xy = (x+y)^n$ and $\frac{dy}{dx} = \frac{y}{x}$ then 3n is	**1)**	2
q)	If the line joining the points (0, 3) and (5, -2) is a tangent to the curve $y = \frac{C}{x+1}$ then the value of C is	**2)**	6
r)	The integral value of b for which the function $f(x) = \left(b^2 - 3b + 2\right)\left(\cos^2 x - \sin^2 x\right) + (b-1)x + \sin(b+1)$ does not posses stationary point	**3)**	9
s)	If $y^3 - y = 2x$ then $y\left[\left(x^3 - \frac{1}{27}\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx}\right]^{-1}$ is	**4)**	4

A) p-1, q-2, r-3, s-4 **B)** p-3, q-4, r-2, s-1

C) p-2, q-4, r-1, s-3 **D)** p-4, q-3, r-2, s-1

54. Match the following

Column I		Column II	
p)	The number of pairs (a,b), a<b of natural numbers such that $\frac{1}{a}+\frac{1}{b}=\frac{1}{2014}$ is	**1)**	9
q)	Let m be the number of 5 element subsets that can be chosen from the set of first 15 natural numbers so that at least two of the five numbers are consecutive. The sum of digits of m is	**2)**	12
r)	If the sum $S=\sum_{n=1}^{2014} n.i^{\left(n^2+n+2\right)}$ is divided by 100 the remainder is	**3)**	13
s)	If a and b are natural numbers such that H.M. of a and b is 2014 then number of pairs (a, b) is	**4)**	15

A) p-1, q-2, r-3, s-4 **B)** p-2, q-3, r-1, s-4
C) p-3, q-2, r-4, s-1 **D)** p-4, q-3, r-2, s-1